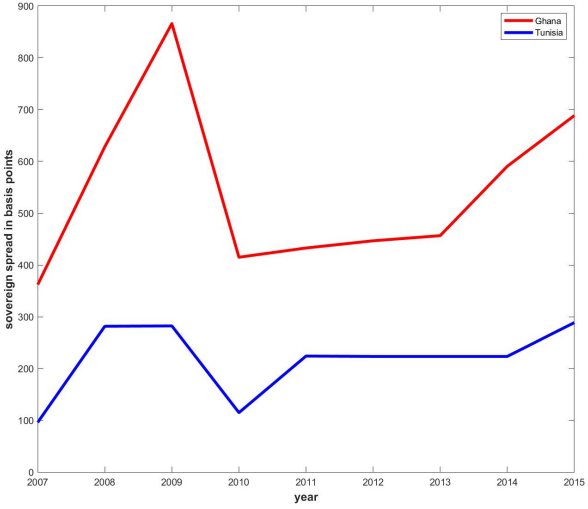
Poverty and Sovereign Default: A Fiscal-Political Amplification Mechanism

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Why Do Poor Countries Pay More to Borrow?



- Tunisia (5% poverty): $\sim$150 bps spread
- Ghana (25% poverty): $\sim$400 bps spread
- **Question:** Why?

Robust Empirical Pattern

International Evidence: 37 emerging economies (1995-2015)

	(3)	(5)	(7)
Poverty proportion	3.16**	3.09**	3.14**
	(1.39)	(1.40)	(1.36)
Debt/GDP	15.87***	16.25***	16.14***
Log dev output	-27.26***	-27.00***	-23.50***
Bribe rate %		19.98**	19.94*

Key Finding

30% higher poverty $\rightarrow$ 120-160 bps higher spreads

Not just a proxy for low income or high inequality

Simple Model Overview

Households:

- Poor (λ): Low income αy
- Non-poor ($1 - \lambda$): Income y
- Transition risk: Prob(p) of becoming poor

Government:

- Taxes non-poor $\rightarrow$ transfers to poor
- Borrows internationally
- Can default strategically

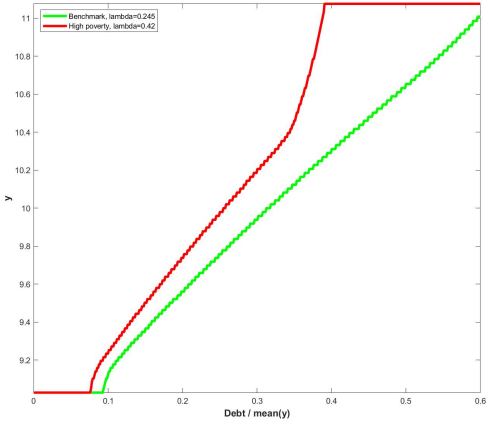
Political Constraint

Non-poor must benefit from participating in fiscal system:
$W_\tau \geq W_0$ (expected utility with insurance vs. without)

How Poverty Amplifies Default Risk

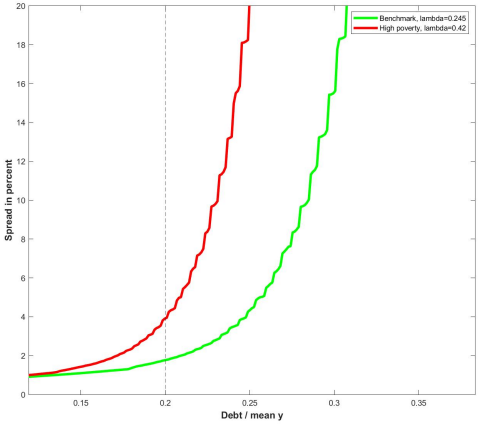
1. **More poverty ($\uparrow \lambda$)**
   - Smaller tax base
   - Larger transfer needs
2. **Two types of defaults emerge:**
   - **Standard defaults:** Low income, high marginal utility
   - **Political defaults:** Non-poor reject fiscal plan (NEW!)
3. **Result:**
   - Default triggered at lower debt levels
   - Lenders demand higher spreads
   - Less borrowing capacity $\rightarrow$ lower welfare

Main Result: Default Regions



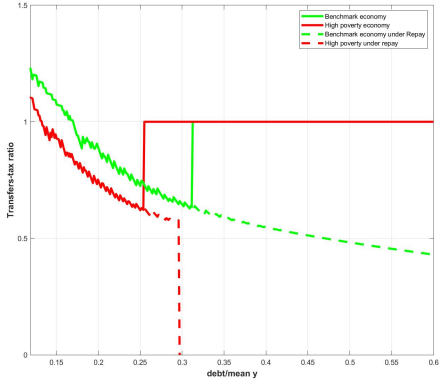
- High-Poverty economy (red) defaults in more states than Benchmark (green)
- For any debt level, needs higher income to avoid default

Higher Spreads for High-Poverty Economy



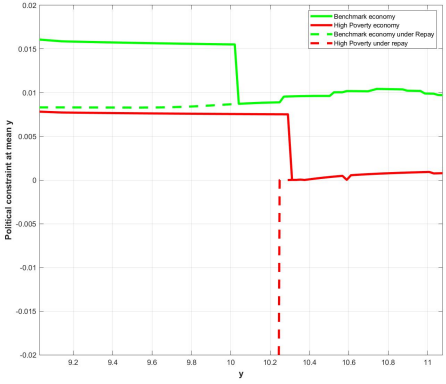
- At any debt level, High-Poverty economy faces higher spreads
- Worse credit terms $\rightarrow$ less borrowing

Mechanism: Transfer-Tax Ratio Falls



- As debt rises, more tax revenue goes to debt service
- Transfer/tax ratio falls below 1
- High-Poverty economy hits crisis point earlier

Political Defaults: A New Mechanism



- Red line (High-Poverty) hits zero $\rightarrow$ **political default**
- Green line (Benchmark) stays positive $\rightarrow$ standard default only
- Political defaults occur to restore fiscal feasibility

Model Matches Data

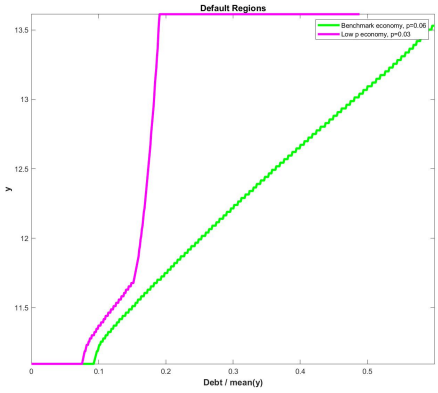
Simulated regression (25 economies with varying λ):

	Data	Model
Poverty coefficient	3.17**	5.69***
	(1.32)	(0.02)
Debt/GDP coefficient	15.82***	21.76***
Log dev output	-25.01***	-33.00***

Success

- Model generates similar magnitude of poverty-spread correlation
- Mechanism is quantitatively relevant

Not Just Inequality



- Yellow line: High-Poverty with inequality controlled
- Effect of poverty remains even when Gini coefficient held constant
- Pure poverty effect is actually stronger

Enhanced Model: Testing Robustness

Question: Is our mechanism an artifact of model simplicity?

Enhanced model adds realism:

1. **Endogenous production:**
   - Labor supply responds to taxes
   - Poor have lower productivity αz
2. **Stone-Geary preferences:**
   - Subsistence constraint $\bar{c}$ for poor
   - Captures absolute nature of poverty
3. **Flexible fiscal policy:**
   - Type-specific tax rates τ_p, τ_n
   - Lump-sum transfers T_p, T_n

Key Addition: Subsistence Constraint

Stone-Geary utility for poor households:

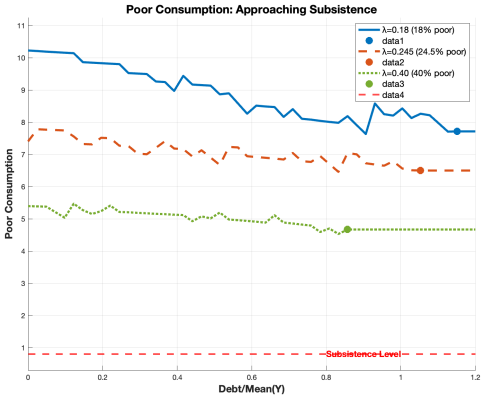
$$U(c_p, l_p) = \frac{\left[(c_p - \bar{c}) - \chi \frac{l_p^{1+\nu}}{1+\nu}\right]^{1-\sigma} - 1}{1-\sigma}$$

- $\bar{c}$ = subsistence level
- As $c_p \to \bar{c}$: marginal utility $\to \infty$
- Creates **hard floor** on government obligations

Key Difference

Poverty is not just low income - it's proximity to survival threshold

Enhanced Model: Example Results



Poor consumption approaches subsistence:

- All economies converge toward $\bar{c}$
- High-poverty hits constraint earlier
- Default prevents falling below

Result

Subsistence constraint accelerates default timing

Enhanced Model Confirms Core Findings

What remains:

- Higher poverty $\rightarrow$ higher spreads
- Political defaults still occur
- Magnitude similar or stronger
- Fiscal-political amplification intact

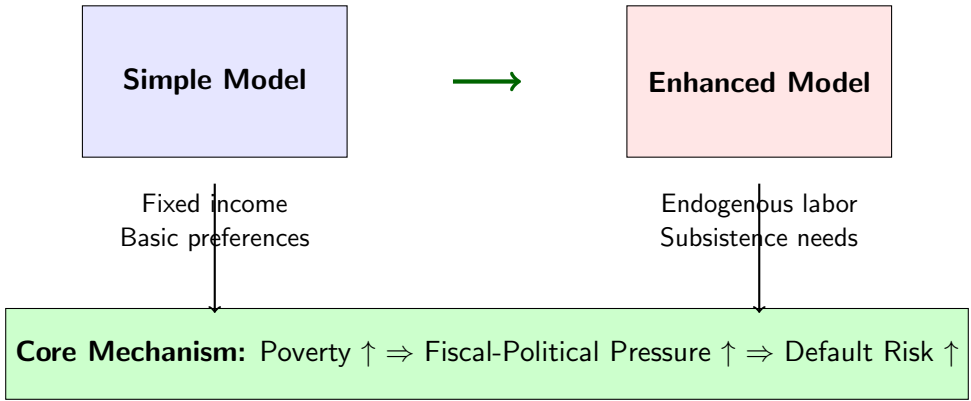
What's new:

- Labor supply falls in crisis
- Subsistence creates urgency
- Endogenous income amplifies
- Richer welfare implications

Bottom Line

Our mechanism is **not** an artifact of simplicity - it's a fundamental feature of poverty and sovereign debt

Visual Summary: Consistent Story



Key Contributions

1. **Novel empirical fact:**
   - Poverty predicts spreads beyond inequality/income
   - Robust across specifications
2. **New theoretical mechanism:**
   - Political defaults when non-poor reject fiscal plan
   - Fiscal-political amplification of default risk
3. **Quantitative success:**
   - Matches data magnitudes
   - 22% more poverty $\rightarrow$ 32% less debt capacity
4. **Robust findings:**
   - Survives model enhancements
   - Not artifact of simplifying assumptions

Policy Implications

1. **Poverty reduction = Financial stability**
   - Not just humanitarian concern
   - Macro-financial benefits quantifiable
2. **Debt relief design**
   - More valuable for high-poverty countries
   - Can break poverty-debt trap
3. **Fiscal institutions matter**
   - Political sustainability as important as fiscal
   - Safety nets must maintain tax compliance
4. **International lending**
   - Consider poverty in risk assessment
   - Concessional terms can be self-sustaining

Thank You!

Questions?

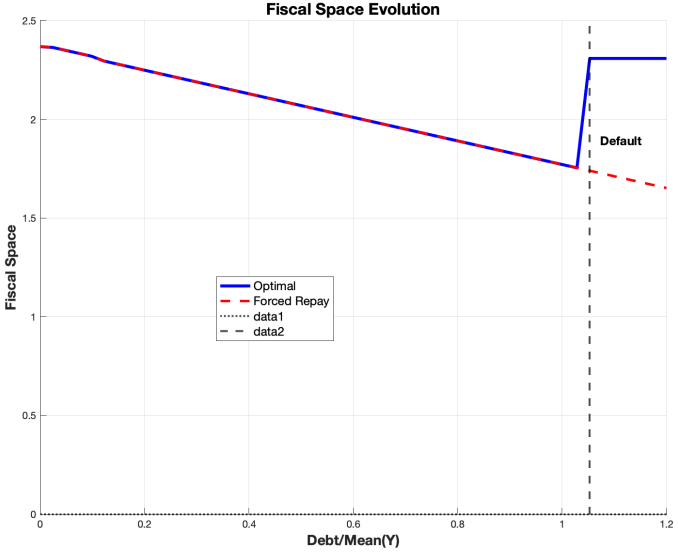
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Backup: Model Calibration

Calibration to South Africa

Parameter	Value	Target/Source
λ	0.245	Poverty headcount ratio
α	0.16	Poor income share
p	0.06	Transition probability (NIDS)
σ	2	Risk aversion
λ_{pol}	0.067	Social expenditure/GDP
β	0.927	Debt/GDP
δ	0.084	Bond duration

Backup: Additional Mechanisms



Backup: Welfare Analysis

